Supporting Information

Mechanism-constrained deep learning recovers individual pharmacokinetics under undernutrition and water-toxicant co-exposure — an in silico study

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Every number reproduces from the accompanying code with `bash scripts/run_all.sh`, followed by the sensitivity commands given in S6.

S1. Accuracy by drug

Held-out patients, random patient split, 5,400 records, against the noise-free simulated curve at all 50 time points.

Drug	log-RMSE	Median fold error	Within twofold
Fluconazole	0.031	1.019	100.0%
Gentamicin	0.049	1.019	100.0%
Paracetamol	0.066	1.039	100.0%
Ceftriaxone	0.079	1.041	100.0%
Lumefantrine	0.159	1.092	99.7%
Isoniazid	0.229	1.080	98.1%
Albendazole	0.248	1.105	97.6%
Amoxicillin	0.277	1.083	96.9%
Rifampicin	0.313	1.114	95.3%

Accuracy is highest for drugs whose disposition is governed by few mechanisms (fluconazole, gentamicin) and lowest for those where absorption and metabolism both move under malnutrition (rifampicin, amoxicillin, albendazole).

S2. Uncertainty calibration

Nominal level	Total interval vs observations	Epistemic vs truth, raw	Epistemic vs truth, recalibrated
50%	49.4%	20.3%	54.0%
80%	79.4%	36.6%	78.3%
90%	89.3%	45.0%	86.0%
95%	94.2%	51.5%	90.5%

Fitted variance-recalibration scale: **3.13**, estimated on the validation split and applied unchanged to test.

S3. Permutation importance, top ten

Each feature permuted within its observed marginal, holding all else fixed; two repeats, held-out records. Laboratory values are permuted jointly with their missingness mask.

Rank	Feature	Block	Δ log-RMSE
1	body weight	anthropometry	0.3893

Rank	Feature	Block	Δ log-RMSE
2	oedema grade	anthropometry	0.2031
3	sex	anthropometry	0.1969
4	water arsenic	toxicants	0.1596
5	serum albumin	labs	0.1572
6	weight-for-height percentage	anthropometry	0.0895
7	weight-for-height z-score	anthropometry	0.0714
8	water lead	toxicants	0.0523
9	body mass index	anthropometry	0.0453
10	C-reactive protein	labs	0.0404

Rank 3 is analysed in S7.

S4. Held-out-drug transfer

Trained without rifampicin and gentamicin; tested on those two only.

Held-out drug	Median fold error	Within twofold
Gentamicin	19.42	8.7%
Rifampicin	4.77	27.2%

S5. Pharmacodynamic target audit

Targets as implemented, with sourcing status. The index scored for `auc_mic` drugs is **free** AUC over the dosing interval divided by MIC.

Drug	Index	MIC (mg/L)	Target	Toxicity metric	Toxic threshold (mg/L)	Source
Amoxicillin	fT>MIC	4.0	0.50	Cmax	60	Craig 1998 (class-level)
Ceftriaxone	fT>MIC	1.0	0.60	Cmax	400	Craig 1998 (class-level)
Gentamicin	Cmax/MIC	1.0	8.0	**Cmin**	**0.5**	Hollander et al. 2023
Lumefantrine	day-7 trough	0.28	0.28	Cmax	25	WWARN 2015 (200–256 ng/mL)
Fluconazole	AUC/MIC	2.0	110	Cmax	80	EUCAST 2008 (fAUC/MIC ≥100)
Rifampicin	AUC/MIC	0.25	38	Cmax	35	**unsourced**
Isoniazid	AUC/MIC	0.05	14	Cmax	12	**unsourced**
Albendazole	AUC/MIC	0.10	2.0	Cmax	8	**unsourced**
Paracetamol	AUC/MIC	5.0	20	Cmax	120	**placeholder, not a PD claim**

Targets tested against a well-nourished reference child

Forty virtual children drawn from the adequate-nutrition stratum, dosed at the label regimen, scored by the simulator. This isolates whether a target is reachable *before* any malnutrition effect.

Drug	Index reached	Implemented target	Attained?
Rifampicin	24.1	38	**no**
Amoxicillin	0.29	0.50	**no**
Isoniazid	232	14	yes, ~16× over
Fluconazole	86.6	110	marginal
Gentamicin	16.5	8.0	yes
Albendazole	8.0	2.0	yes
Ceftriaxone	1.00	0.60	yes
Lumefantrine	3.08	0.28	yes

Rifampicin and amoxicillin do not reach their implemented targets in a well-nourished child, so the low baseline attainment reported for them in the main text is not attributable to undernutrition. Isoniazid's target is met by a wide margin and published AUC0-24/MIC targets for isoniazid are 567 (murine) and ~707 (in vitro), roughly fortyfold higher than the implemented value.

S6. Target sensitivity analyses

Gentamicin trough threshold

The only toxicity signal in the dosing experiment.

Threshold	Source	Over-exposed, label	Over-exposed, model	Attainment
2.0 mg/L	unsourced (previous implementation)	0.0%	0.0%	100% → 100%
1.0 mg/L	neonatal target	2.5%	0.0%	100% → 100%
0.5 mg/L	**paediatric target, implemented**	**12.5%**	**0.0%**	100% → 100%

```
python scripts/clinical_impact.py --run runs/main_hybrid --n 120 \
    --drugs gentamicin --policy anchored
```

with `toxic_threshold` set to each value in `malnupk/drugs.py`. Outputs in `outputs/ci_gent_*.json`.

Amoxicillin fT>MIC target

Target	Attainment, label	Attainment, model	Median dose
50% (implemented)	7.5%	15.0%	2.16×
40%	19.2%	71.7%	1.85×

Output in `outputs/ci_amox_target40.json`.

S7. The sex over-attribution diagnostic

Sex enters the simulator at one point only: the sex-specific constants of the fat-free-mass relation, which propagate to total body water and hence to distribution volume. Its causal magnitude is therefore directly measurable.

Procedure. Sample 400 virtual patients. For each, flip `male`, rebuild the fat-free mass and body-water compartments that follow from it, and re-derive the pharmacokinetic parameters for all nine drugs. Anthropometry (age, weight, height, MUAC, oedema grade) is held fixed, so only the body-composition pathway moves. The model's response is obtained by flipping the same input in the held-out feature matrix and comparing predicted parameters.

Parameter	Simulator (RMS log change)	Model (RMS log change)	Ratio
CL	0.0000	0.0220	model responds; simulator does not
V1	0.0422	0.1476	3.5×
Q	0.0000	0.0196	model responds; simulator does not
V2	0.0417	0.1983	4.8×
ka	0.0000	0.0341	model responds; simulator does not
F	0.0000	0.0234	model responds; simulator does not
fu	0.0000	0.0307	model responds; simulator does not

Flipping sex changes fat-free mass by a median of 3.3% at identical weight, height and oedema grade.

Interpretation. The model moves the distribution volumes three-and-a-half to nearly five times more than the simulator does, and moves five parameters the simulator does not move at all. The permutation importance of sex is therefore a true statement about the model and a false one about the mechanism.

Most likely cause. Four other inputs — height-for-age, weight-for-height and weight-for-age z-scores, and weight-for-height percentage — are computed against sex-specific reference curves, so sex is partly recoverable from them. With redundant inputs, a network can distribute a learned effect among them in any proportion. The seven-parameter bottleneck constrains what the intermediate quantities mean, not how the encoder arrives at them, and auxiliary supervision on latent physiology did not prevent the over-attribution.

Reproduce with `python scripts/diagnose_sex.py`.

S8. Dosing results in full

120 virtual patients disjoint from all training and validation data, 8 drugs, 960 decisions. Anchored policy. Both arms scored by the simulator.

By drug

Drug	Reached target	Exceeded toxicity	Median dose ratio	Interval changed
Rifampicin	8.3% → 87.5%	0.0% → 0.0%	1.85×	4%
Albendazole	62.5% → 98.3%	0.0% → 0.0%	1.17×	6%
Isoniazid	74.2% → 99.2%	0.0% → 0.0%	1.17×	0%

Drug	Reached target	Exceeded toxicity	Median dose ratio	Interval changed
Amoxicillin	7.5% → 15.0%	0.0% → 0.0%	2.16×	0%
Gentamicin	100% → 100%	12.5% → 0.0%	1.00×	6%
Lumefantrine	100% → 100%	0.0% → 0.0%	1.00×	0%
Ceftriaxone	100% → 100%	0.0% → 0.0%	1.00×	0%
Fluconazole	100% → 100%	0.0% → 0.0%	1.00×	0%

By nutritional stratum

Stratum	Reached target	Dose ratio, 10th–90th percentile
Adequate	73.6% → 86.6%	1.00–1.85×
Mild	71.9% → 87.5%	1.00–1.85×
Moderate	71.0% → 87.0%	1.00–1.85×
Severe, marasmic	65.0% → 87.2%	1.00–2.16×
Severe, oedematous	69.1% → 90.8%	1.00–4.00×

Policy comparison

Policy	Reached target	Exceeded toxicity	Median dose
Label (weight band)	69.1%	1.6%	1.00×
Anchored (default)	87.5%	0.0%	1.00×
Minimum effective	87.3%	0.0%	0.63×

S9. Claims without a verified source

Nine assertions in the Introduction or Methods of the main manuscript have no verified primary source. They are reproduced here for convenience; the main text carries the same list.

Claim	Status
Visceral organ mass falls sub-proportionally to body weight (the 0.45 sparing exponent)	No measured source.
Extracellular water expands with reciprocal intracellular depletion	Total body water elevation is sourced; the compartment split is not.
Creatinine-based GFR is biased upward at low muscle mass	The paediatric equation is sourced; the bias is not.
EED specifics: villous blunting, raised gastric pH, bile-salt deficiency	EED in general is sourced; these three mechanisms are not.
AGP rise lowers unbound fraction of basic drugs	The acute-phase rise and basic-drug binding are sourced; the combined claim is an inference.
DMT1 up-regulation extends to lead and manganese, and to children	Sourced for cadmium in rats.
Arsenic methylation is methyl-donor limited under protein deficit	SAM dependence is sourced; nutritional limitation is not.
Cadmium nephrotoxicity reduces glomerular filtration	Tubular dysfunction is sourced at borderline significance; filtration was not measured.

Claim	Status
PK/PD targets for rifampicin, isoniazid, albendazole; paracetamol placeholder	See S5.

For five of these the general pharmacology is sourced and the nutrition-dependent limb is not. That limb is the coupled nutrition–toxicant axis, which is the distinctive claim of the simulator, and it currently rests on mechanistic plausibility rather than on measurement.

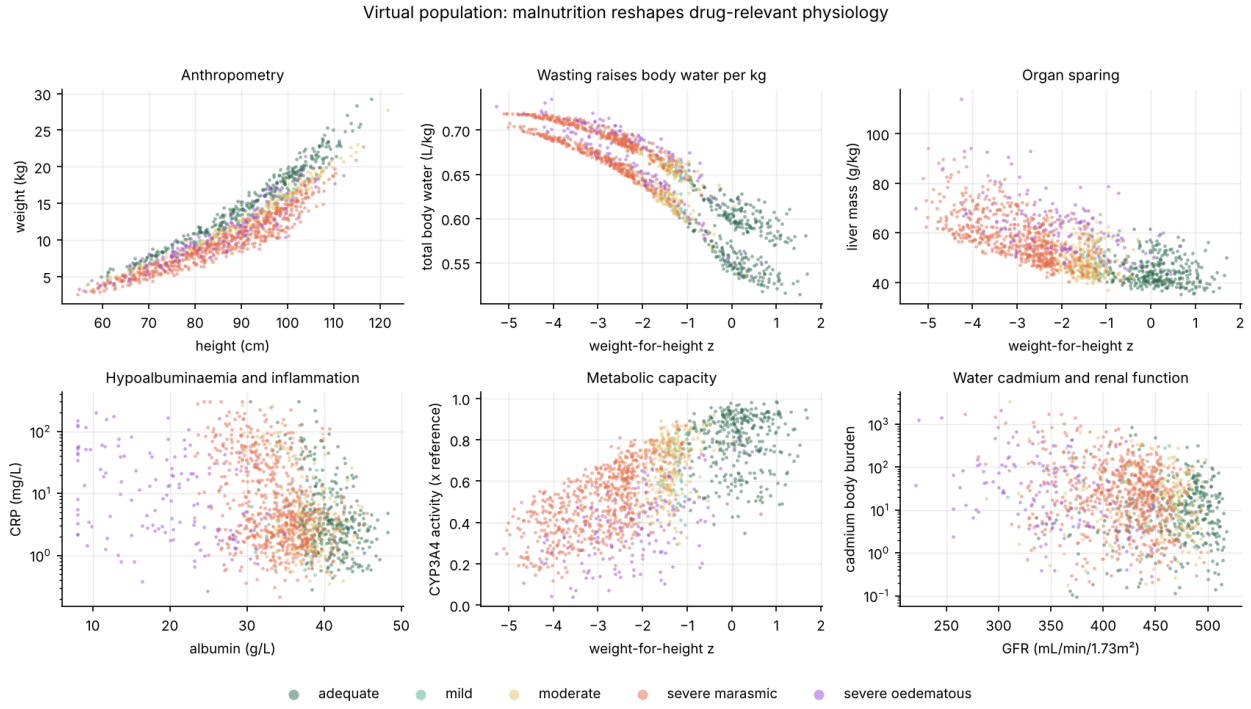


Figure S1 | The simulated population. Anthropometry, body composition, organ mass and derived physiology across the five nutritional strata.

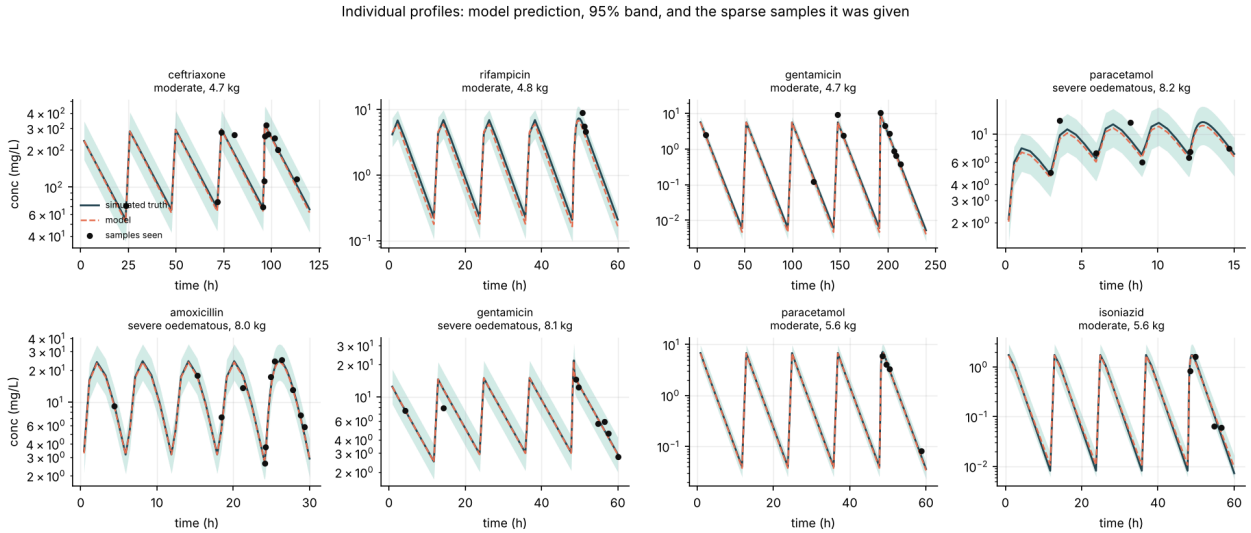


Figure S2 | Representative predicted and simulated concentration–time profiles, across drugs and nutritional strata, showing the sparse observed samples the model actually received.